Software Requirement Specs

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Srs Content

Software Requirements Specification (SRS) for University Class Scheduler (UCS)

1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the required capabilities, constraints, and external interfaces of the University Class Scheduler (UCS), a platform-independent, AI-driven scheduling system for higher education institutions. Its purpose is to serve as the foundational agreement between stakeholders—including Department Staff, University Administrators, Catalogue Creators, Webmasters, Students, and Professors—and the development team. It provides a complete, unambiguous, and verifiable description of the system to guide design, implementation, verification, and validation activities in accordance with IEEE 830-2013 and ISO/IEC/IEEE 29148:2018.

1.2 Scope

The UCS replaces legacy manual and static scheduling processes with an autonomous, real-time, and predictive platform. In scope are: (a) the Core Optimization Engine for conflict-free schedule generation balancing room capacity, departmental proximity, faculty commute time, workload enforcement, and well-being constraints; (b) role-specific interfaces including a web-based Department Staff Dashboard, Executive KPI Dashboard, Catalogue Editor, Admin Tools, Student Mobile App, and Professor Interface; (c) the Scenario Sandbox for predictive bottleneck simulation and 'what-if' policy impact forecasting; (d) real-time, SSO-authenticated synchronization across mobile and web platforms; (e) GPS-enabled indoor campus navigation with offline map caching; (f) FERPA-compliant data handling, WCAG 2.1 AA accessibility, Azure GovCloud deployment, and integration with Banner, Outlook/Exchange, and Oracle 11g without requiring their upgrade. Out of scope are student enrollment/registration functionality, LMS or gradebook integration, physical IoT room sensors, HR/payroll systems, external accreditation reporting, and native desktop applications.

1.3 Definitions, Acronyms, and Abbreviations

- UCS: University Class Scheduler
- SIS: Student Information System
- SSO: Single Sign-On (Shibboleth or Azure AD)

- FERPA: Family Educational Rights and Privacy Act
- WCAG: Web Content Accessibility Guidelines
- API: Application Programming Interface
- GPS: Global Positioning System
- UI: User Interface
- UX: User Experience
- KPI: Key Performance Indicator
- SLA: Service Level Agreement
- CI/CD: Continuous Integration / Continuous Deployment
- PII: Personally Identifiable Information
- TLS: Transport Layer Security
- AES: Advanced Encryption Standard
- schedule: noun only — a structured timetable of course sections, instructors, rooms, and times (added per terminology sweep)
- sandbox: interactive, isolated environment for simulating scheduling changes without affecting production data
- soft constraint: a non-mandatory scheduling rule that contributes to optimization scoring but does not invalidate a schedule if violated (e.g., departmental co-location, commute minimization)
- permeability: measure of how easily faculty or students can move between adjacent academic buildings, calculated from walkway topology and door access policies
- building completeness: percentage of indoor floor plans fully mapped and validated for GPS-assisted navigation

1.4 References

- IEEE Std 830-2013, "Recommended Practice for Software Requirements Specifications"
- ISO/IEC/IEEE 29148:2018, "Systems and software engineering — Life cycle processes — Requirements engineering"
- FERPA Regulations (34 CFR Part 99)
- WCAG 2.1 AA Standards (W3C Recommendation, June 2018)
- Azure GovCloud Compliance Documentation (Microsoft)
- University IT Infrastructure Policies v3.2
- Operational Environment List (OEL), artifact ID: operating_env_list_Deployer_v1.0
- System Requirements List (SRL), artifact ID: system_requirements_Analyst_v1.0
- Requirement Model (RM), artifact ID: requirements_model_Analyst_v1.0
- Business Requirement Document (BRD)
- Use Case Diagram (see Requirement Model)
- Data Flow Diagram (DFD)
- Entity-Relationship Diagram (ERD)
- Data Dictionary
- Logical Architecture Diagram (annexed separately)
- User Workflow for Schedule Generation (annexed separately)

2 Overall Description

2.1 Product Perspective

UCS is a standalone, cloud-native application that integrates into the university's existing digital ecosystem. It does not replace upstream systems (Banner SIS, Outlook/Exchange, Oracle 11g) but consumes their data via secure APIs and feeds validated outputs back where appropriate. UCS acts as a scheduling orchestration layer—accepting academic metadata from the

Catalogue Editor, applying optimization logic, exposing results through role-specific interfaces, and enabling scenario-based analysis before production commitment. It is deployed exclusively on Azure GovCloud and operates behind the university's SSO infrastructure.

2.2 Product Functions

The system shall provide the following high-level functions:

- Autonomous generation of conflict-free course schedules for 800–1,200 sections per college within ≤18 minutes.
- Secure, role-based web dashboard for Department Staff to review, override, and approve schedules.
- Interactive Scenario Sandbox enabling Department Staff and Administrators to simulate scheduling changes and receive pre-emptive bottleneck diagnostics and resolution options.
- Real-time (<5 sec latency) bi-directional schedule synchronization for Students and Professors across iOS and Android mobile apps and responsive web interfaces.
- GPS-enabled indoor campus navigation with turn-by-turn walking directions, live transit delay adjustments, and offline-capable cached map tiles.
- Faculty commute optimization engine that flags high-friction pairs (>10 min walk, <30 min break) and delivers push notifications with one-tap request for alternatives.
- Structured, version-controlled, and auditable interface for Catalogue Creators to edit and validate course metadata against constraint engine rules.
- Executive dashboard for University Administrators displaying real-time KPIs (room utilization, departmental clustering %, faculty high-friction commute pairs, cross-college optimization comparisons).
- Drag-and-drop class swap functionality for Students with immediate contextual feedback on eligibility, conflicts, and AI-suggested resolutions.
- Comprehensive audit logging capturing actor, action, timestamp, target entity, device/IP, and outcome for all user-initiated and system-triggered events.

2.3 User Classes and Characteristics

Six distinct user classes have been identified, each with unique goals, responsibilities, technical proficiency, and interaction patterns:

- Department Staff: Mid-level academic operations personnel responsible for logistics and faculty coordination; moderate technical skill; interact primarily via the web dashboard to review, override, and approve schedules; require authoritative control over constraint exceptions and scenario recommendations.
- University Administrator: Senior strategic decision-maker overseeing resource allocation and policy compliance; low-to-moderate technical proficiency; interacts via the executive dashboard to monitor KPIs and assess cross-departmental optimization outcomes.
- University Catalogue Creators: Academic domain experts defining course metadata; detail-oriented; use a structured web editor to input prerequisites, attributes, and credit hours; require version control, automated validation, and constraint engine integration testing.
- University Webmaster: Technical infrastructure specialist ensuring alignment with identity federation, accessibility standards, and deployment pipelines; highly technical; manages SSO bridges, WCAG audits, and CI/CD integrations via admin tools.
- Students: Primary end-users seeking personal schedule access and wayfinding; highly mobile-centric; interact exclusively via intuitive iOS/Android apps featuring drag-and-drop swaps, real-time sync, and indoor GPS navigation.
- Professors: Academic staff requiring contextual teaching assignments and well-being safeguards; time-constrained and privacy-conscious; use mobile and web interfaces to view personalized schedules, receive commute alerts, and submit optimization feedback.

2.4 Operating Environment

UCS operates in a hybrid environment spanning cloud infrastructure, web browsers, and mobile devices:

- Cloud Infrastructure: Azure GovCloud only; containerized backend services deployed via university-approved CI/CD pipelines; compute nodes optimized for CPU-intensive optimization workloads (e.g., Azure Dv5/Ev5 series); storage encrypted at rest using AES-256.
- Web Dashboard: Accessible via modern browsers—Chrome, Firefox, Safari, and Edge (latest two stable versions); supports keyboard navigation, screen readers, and voice-guided readiness per WCAG 2.1 AA.
- Mobile Applications: Distributed via Apple App Store and Google Play; requires iOS 15+ and Android 11+; supports offline mode for core schedule viewing and cached indoor maps.
- Staff Tablet Dashboard: Minimum 8-inch screen; modern browser; stable Wi-Fi connectivity.
- Network: HTTPS required for all traffic; real-time sync requires <5 sec latency; offline mode must function without network connectivity for core schedule and map functions; campus-wide Wi-Fi coverage required for indoor positioning accuracy and live transit delay adjustments.

2.5 System Constraints

- Deployment: Must be hosted exclusively on Azure GovCloud; no hybrid or on-premise hosting permitted.
- Authentication: All users must authenticate via university SSO (Shibboleth or Azure AD); no local credential storage or custom authentication logic permitted.
- Legacy Integration: Must interoperate with Banner (SIS), Outlook/Exchange, and Oracle 11g (room booking) without requiring upgrades to those systems.
- Accessibility: All interfaces must comply with WCAG 2.1 AA standards; verified via automated scanning and manual expert audit.
- Data Handling: All student-identifiable data must be treated as FERPA-protected; no persistent local caching of such data beyond 24 hours on mobile devices.
- Map Rendering: Indoor positioning and offline tile caching must be supported for all key academic buildings.
- Protocol: Real-time synchronization must use WebSocket or equivalent low-latency protocol.

2.6 Assumptions and Dependencies

- Assumptions:
- University SSO infrastructure remains available and responsive during peak usage periods.
- Banner, Outlook/Exchange, and Oracle 11g APIs remain stable and accessible per documented SLAs.
- Campus Wi-Fi coverage is sufficient to support indoor positioning accuracy and live transit updates.
- Faculty and students possess smartphones meeting OS version requirements and grant location permissions.
- Department Staff will perform final human review and approval prior to schedule publication.
- Third-Party Dependencies:
- Azure GovCloud
- University SSO (Shibboleth/Azure AD)
- Banner SIS: RESTful API v3.2 (OAuth2.0 bearer tokens, read-only access to course, section, instructor, and enrollment data; fallback: CSV batch export via SFTP if API unavailable for >15 min)
- Outlook/Exchange: Microsoft Graph API v1.0 (free/busy calendar reads; write access limited to UCS-managed event creation/modification; fallback: polling every 5 min with cached state if Graph API degrades)
- Oracle 11g Room Booking: JDBC connector v2.1 (read-only; uses university-provided thin client; fallback: static weekly room capacity snapshot if connection fails >30 min)
- Google Maps Platform: Maps SDK for Android/iOS and Maps JavaScript API for web; indoor mapping enabled via custom floor plans and POI metadata; fallback: vector-based offline map renderer using OpenStreetMap-derived building footprints if Maps API exceeds quota)
- FCM (Firebase Cloud Messaging) / APNs (Apple Push Notification service)

3 Specific Requirements

3.1 System Features

F-01: Academic Scheduling Engine
The system shall autonomously generate, validate, and optimize course schedules using AI-driven constraint programming and multi-objective scoring. This feature encompasses FR-001 (Autonomous Schedule Generation), FR-002 (Department Staff Schedule Review & Approval), and FR-003 (Scenario Sandbox Simulation). It enables deterministic, repeatable, and auditable scheduling outcomes aligned with academic policy.

F-02: Real-Time Synchronization & Mobility
The system shall deliver consistent, low-latency, and secure access to personal schedules across all user devices and contexts. This feature encompasses FR-004 (Real-Time Schedule Synchronization), FR-005 (Indoor Campus Navigation), FR-006 (Faculty Commute Optimization & Alerts), FR-009 (Student Class Swap Functionality), and FR-010 (Offline Mobile Functionality). It ensures continuity of experience regardless of connectivity or device type.

F-03: Academic Metadata Governance
The system shall provide a version-controlled, rule-governed, and auditable workflow for managing course catalogue metadata. This feature encompasses FR-007 (Catalogue Metadata Management) and supports integrity, traceability, and regulatory compliance of academic definitions.

F-04: Strategic Decision Support
The system shall provide actionable, real-time analytics and KPIs to inform institutional resource planning and policy evaluation. This feature encompasses FR-008 (Executive KPI Dashboard) and links operational scheduling data to strategic objectives.

3.2 Functional Requirements

FR-001: Autonomous Schedule Generation

The system shall generate a conflict-free course schedule using an AI-driven optimization engine that balances room capacity, departmental proximity, faculty commute time, workload enforcement, and well-being constraints without manual intervention. Acceptance criteria: (a) For a test dataset of 1,000 sections, the generated schedule must contain zero hard constraint violations (e.g., room over-capacity, time conflicts, prerequisite mismatches); (b) At least 90% of faculty assignments must satisfy soft constraints (commute minimization, departmental co-location) as measured by weighted objective scoring using historical conflict data ≥2 academic years; (c) Output must include a machine-readable validation report detailing constraint adherence metrics. Verification method: Automated acceptance test suite executed in staging environment using synthetic dataset matching production scale; pass/fail determined by parsing JSON validation report and validating counts against thresholds. Pass criterion: 100% zero hard violations, ≥90% soft constraint satisfaction, and valid JSON report structure. Priority: High.

Priority: High.

FR-002: Department Staff Schedule Review & Approval

The system shall provide a secure, web-based dashboard for Department Staff to review auto-generated schedules, inspect adherence to departmental constraints, override specific assignments, and formally approve or request revisions before finalization. Acceptance criteria: (a) Dashboard must display side-by-side comparison of proposed vs. prior term schedule; (b) Override controls must allow modification of instructor, room, or time slot for any section with mandatory reason code entry and audit trail capture; (c) Formal approval must trigger a signed digital certificate and initiate downstream publishing workflow. Verification method: Manual usability test with 5 Department Staff representatives using test accounts in UAT environment; success confirmed when all three criteria observed and logged. Pass criterion: 100% of participants successfully completing all three tasks. Priority: High.

Priority: High.

FR-003: Scenario Sandbox Simulation

The system shall provide an interactive 'Scenario Sandbox' enabling Department Staff and University Administrators to simulate scheduling changes and receive pre-emptive identification and resolution options for logistical bottlenecks including room conflicts, faculty overload, and proximity violations. Acceptance criteria: (a) Users must be able to configure parameters (e.g., add 5 new lab sections, relocate Physics department to Building X) and execute simulation in <90 seconds; (b) Bottleneck reports must categorize issues by type, severity, root cause, and include ≥2 actionable resolution options per high-severity finding; (c) All sandbox activity must be isolated from production data and require explicit approval before influencing live output. Verification method: Automated performance and correctness test in staging: inject 10 predefined scenarios; measure execution time and validate report structure and option count via schema validation (JSON Schema v7) and content extraction. Pass criterion: 100% of simulations complete in <90 s; 100% of high-severity findings include ≥2 resolution options; 100% of sandbox sessions operate in isolated transaction context. Priority: High.

Priority: High.

FR-004: Real-Time Schedule Synchronization

The system shall deliver real-time (<5 sec latency, 95th percentile) synchronization of personal schedules for Students and Professors across iOS/Android apps via SSO-authenticated, encrypted channels in Online Sync Mode. Acceptance criteria: (a) A change committed on mobile app must appear on all registered devices (mobile + web) within 5 seconds at 95th percentile; (b) Sync failure must trigger automatic retry with exponential backoff and local conflict resolution log; (c) All sync payloads must be signed and encrypted end-to-end using TLS 1.2+. Verification method: Load test using Gatling simulation (100 concurrent users) measuring end-to-end latency distribution; failure injection test verifying retry behavior and log persistence in database. Pass criterion: 95th percentile latency ≤5.0 s; 100% of failed sync attempts trigger retries; 100% of logs persisted and timestamped. Priority: High.

Priority: High.

FR-005: Indoor Campus Navigation

The mobile app shall render turn-by-turn walking directions from a user's current GPS location to scheduled classrooms, including indoor map overlays, live transit delay adjustments, and offline-capable cached map tiles for key buildings. Acceptance criteria: (a) Navigation must initialize and display first route segment within 1.2 seconds (95th percentile) on device; (b) Indoor maps must load and render fully for ≥95% of designated academic buildings when offline; (c) Transit delay adjustments must update route dynamically upon receipt of live feed from campus transit API. Verification method: Instrumented iOS/Android test harness measuring cold-start and offline rendering times on reference devices (iPhone 12, Pixel 5); validation against pre-cached building list and transit API mock responses. Pass criterion: 95th percentile cold-start ≤1.2 s; pixel-diff validation confirms ≥95% of floor plans rendered correctly offline; 100% of simulated delays trigger route recalculation. Priority: High.

Priority: High.

FR-006: Faculty Commute Optimization & Alerts

The system shall minimize faculty commute time between back-to-back classes and flag high-friction scheduling pairs (e.g., >10 min walk, <30 min break); Professors shall receive push notifications with well-being alerts and one-tap options to request optimized alternatives. Acceptance criteria: (a) ≥95% of identified high-friction pairs must trigger a push notification within 30 seconds of schedule generation or change; (b) One-tap 'Request Alternative' must return ≥1 valid alternative satisfying all hard constraints and improving commute time by ≥25% within 60 seconds; (c) Alert messages must include estimated walking time, break duration, and distance. Verification method: End-to-end test simulating schedule publish event and measuring notification delivery latency (via FCM/APNs logs); subsequent API call to /commute/alternatives endpoint timed and result validated against constraint solver output. Pass criterion: ≥95% notification delivery rate within 30 s; 100% of /commute/alternatives calls return ≥1 valid alternative within 60 s; 100% of alerts contain all three required fields. Priority: High.

Priority: High.

FR-007: Catalogue Metadata Management

The system shall provide a structured web interface for University Catalogue Creators to edit, version-control, and validate course metadata, with auditable, reversible changes and automatic constraint engine validation before production deployment.

Acceptance criteria: (a) Each catalogue edit must generate a new immutable version with timestamp, author, and diff summary; (b) Prerequisite chain validation must detect cycles and attribute dependency failures before save; (c) Version rollback must restore all associated constraint engine rules and metadata without side effects. Verification method: Automated functional test suite executing version create → save → diff → rollback sequence in staging; validation via database snapshot comparison and constraint engine re-execution. Pass criterion: 100% of versions created with full audit metadata; 100% of invalid prerequisite chains rejected before save; 100% of rollbacks restore original state without side effects. Priority: Medium.

Priority: Medium.

FR-008: Executive KPI Dashboard

The system shall provide an executive dashboard for University Administrators displaying real-time KPIs—including room utilization rate, departmental clustering %, faculty high-friction commute pairs, and cross-college optimization comparisons—to inform strategic resource allocation and policy decisions. Acceptance criteria: (a) Dashboard must render all KPI visualizations within 2 seconds on initial load; (b) Cross-college comparison report must export to PDF/CSV with <5% statistical variance from source data warehouse; (c) KPIs must update in near real-time (≤2 min lag) from underlying schedule database. Verification method: Frontend performance test (Lighthouse v10) measuring TTI and FCP; export test comparing checksums and sampling 100 rows from exported CSV against warehouse query results; lag test measuring timestamp delta between DB commit and dashboard refresh. Pass criterion: 100% of visualizations load ≤2.0 s; statistical variance <5% across 100 sampled rows; lag ≤2.0 min for 100% of KPIs. Priority: High.

Priority: High.

FR-009: Student Class Swap Functionality

The system shall enable students to drag-and-drop class swaps within the mobile app and receive immediate, contextual feedback on academic eligibility, room availability, time conflicts, and AI-suggested resolution options. Acceptance criteria: (a) Swap attempt must return validation result within 800 ms (95th percentile); (b) Feedback must distinguish hard failures (e.g., prerequisite gap) from soft suggestions (e.g., 'Lab section L2 has higher utilization; consider L1'); (c) Confirmed swaps must persist across all devices within 5 seconds. Verification method: Mobile instrumentation test measuring API response time and sync propagation; UI automation verifying message classification and persistence. Pass criterion: 95th percentile response ≤800 ms; 100% of messages correctly classified; 100% of swaps propagated within 5 s. Priority: High.

Priority: High.

FR-010: Offline Mobile Functionality

The system shall support core functionality in Offline Cache Mode—including viewing next three scheduled locations, cached indoor map tiles for those buildings, and cached instructor names—without blank screens or authentication prompts. Acceptance criteria: (a) App must launch and display cached schedule within 1 second offline; (b) Indoor map tiles for top 20 most-frequented buildings must be pre-cached and render fully offline; (c) No network-dependent UI elements (e.g., loading spinners, auth redirects) may appear during offline operation. Verification method: Offline test on physical devices (iOS 15+, Android 11+) using airplane mode; timing measurements and visual inspection for UI anomalies. Pass criterion: Launch and display ≤1.0 s; pixel-diff validation confirms 100% of top-20 building floor plans render offline; zero network-dependent UI elements observed. Priority: High.

Priority: High.

3.3 Transition & Deployment Requirements

TR-001: Phased Rollout
The system shall be deployed in three phases: (a) Pilot: 2 colleges, 3 months, limited user group (Department Staff + Catalogue Creators only); (b) Staged Release: 5 additional colleges, 2-month ramp-up with parallel run and reconciliation; (c) Full Production: University-wide cutover during summer break. Verification method: Deployment checklist signed off by PM, Lead Architect, and IT Operations; phase completion confirmed by user adoption metrics and reconciliation report. Pass criterion: 100% of checklist items signed; reconciliation report shows <0.1% error rate and ≥99.99% hash match. Priority: High.

TR-002: Data Migration
All historical schedule data (past 3 academic years) and active course catalogue metadata shall be migrated from Banner SIS and legacy spreadsheets into UCS using idempotent, auditable ETL jobs. Each migration batch shall produce a reconciliation report comparing source row counts, key field hashes, and error logs. Verification method: Reconciliation report review and spot-check validation by Data Steward; successful migration confirmed when error rate <0.1% and hash match ≥99.99%. Priority: High.

TR-003: User Training & Enablement
Role-specific training materials (video, quick-reference guides, sandbox environment access) and live virtual workshops shall be delivered to all user classes prior to Pilot launch. Completion rate ≥90% and post-training survey NPS ≥75 required before Phase 2. Verification method: LMS completion reports and survey analytics exported and reviewed by Change Manager. Priority: High.

TR-004: Cutover & Rollback
System cutover shall occur during a 4-hour maintenance window with 72-hour advance notice. A fully tested rollback procedure (revert to Banner-managed schedule + restore last UCS backup) must be executable within 30 minutes. Success metric: First cutover achieves ≤15 min downtime and zero data loss. Verification method: Dry-run cutover exercise documented and approved by Change Advisory Board. Priority: High.

TR-005: Operational Readiness
UCS shall achieve operational readiness sign-off only after passing all acceptance tests, completing security audit (PenTest + SOC 2 Type II evidence), and validating integration health monitors (Banner API uptime ≥99.9%, Outlook Graph latency <1s p95, Oracle connector response <2s). Verification method: Operational Readiness Certificate signed by CIO, CISO, and VP Academic Affairs. Priority: High.

3.4 Non-Functional Requirements

NFR-001: Performance

- Optimization Engine: Must generate and validate a full draft schedule for 800–1,200 sections in ≤18 minutes under peak registration load (99th percentile). Measured from job submission to validated output delivery.
- Mobile App Responsiveness: 'Refresh Schedule' action must render updated map + walking directions in ≤1.2 seconds (95th percentile) on iOS 15+/Android 11+ devices with ≥2 GB RAM.
- Scenario Sandbox: Must support up to 35 concurrent sessions without queuing or degradation in response time (≤2 sec median latency) or accuracy (≥99.5% correct bottleneck classification).
- Web Dashboard: Page load time for schedule review view must be ≤1.5 seconds (95th percentile) on broadband connections.

NFR-002: Security

- Data Protection: All student-identifiable data must be encrypted at rest using AES-256 and in transit using TLS 1.2+.
- Authentication: All API calls to Banner must route exclusively through university SSO and OAuth2.0; no direct credentials or tokens stored client-side.
- Audit Logging: Full audit logs must capture actor (user_id), action (e.g., 'approve_schedule'), timestamp (ISO 8601), device/IP, and target entity (e.g., 'Schedule.section_id=4567') for all user-initiated and critical system actions.
- FERPA Compliance: No persistent local caching of student-identifiable data on mobile devices beyond 24 hours; cache purge must occur automatically on app exit or after idle timeout.
- Role-Based Access Control: Permissions must align strictly with university identity groups (e.g., 'dept_staff@university.edu' → Department Staff role); no privilege escalation paths permitted.

NFR-003: Usability & Accessibility

- WCAG 2.1 AA Compliance: All interfaces must pass automated scanning (axe-core v4+) and manual expert audit for all Level A and AA success criteria, including full keyboard navigation, ARIA labels, sufficient color contrast (≥4.5:1), screen reader compatibility, and voice-guided navigation readiness.
- Mobile-First Design: All core workflows (schedule view, class swap, navigation) must be fully functional and intuitive on iOS/Android touch interfaces without zooming or horizontal scrolling.

- Cognitive Load: Critical alerts (e.g., well-being warnings) must use standardized iconography, plain language, and progressive disclosure; no more than three primary actions visible at once.

NFR-004: Reliability

- Uptime: System must achieve ≥99.5% uptime during academic term windows (excluding scheduled maintenance windows of ≤2 hrs/month with 72-hr notice).
- Failover: All critical services (optimization engine, sync broker, notification service) must support automatic failover to redundant instances within ≤30 seconds.
- Data Integrity: Schedule generation must be transactional; partial writes or inconsistent states must be rolled back with atomic error reporting.
- Recovery Point Objective (RPO): Maximum data loss tolerance is 5 seconds for real-time sync events.
- Recovery Time Objective (RTO): Critical service restoration must occur within 15 minutes of incident detection.

4 External Interface Requirements

4.1 User Interfaces

- Web Dashboard: Responsive HTML5/CSS3/JavaScript interface supporting Chrome, Firefox, Safari, Edge (latest two versions); WCAG 2.1 AA compliant; keyboard navigable; screen reader tested; minimum viewport width 1024px.
- Mobile Apps: Native iOS (Swift) and Android (Kotlin) applications distributed via official app stores; follow platform-specific Human Interface Guidelines (HIG) and Material Design principles; support dynamic type scaling and dark mode.
- Staff Tablet Dashboard: Optimized for 8–10 inch tablets; touch-first layout; offline-capable for schedule review; integrates with university MDM for security policy enforcement.

4.2 Hardware Interfaces

- Mobile Clients: Must support iOS 15+ and Android 11+; require GPS hardware and location services enabled; support Bluetooth LE for optional indoor beacon augmentation.
- Backend Infrastructure: Azure GovCloud Dv5/Ev5-series VMs or AKS clusters with GPU acceleration optional for future ML model inference; SSD-backed storage; dedicated network bandwidth for optimization jobs.

4.3 Software Interfaces

- Identity Federation: Integration with university Shibboleth or Azure AD via SAML 2.0 or OpenID Connect; no local user database maintained.
- Banner SIS: RESTful API integration using OAuth2.0 bearer tokens issued by university SSO; read-only access to course, section, instructor, and student enrollment data.
- Outlook/Exchange: Graph API integration to read faculty calendar free/busy slots; write access to UCS-managed schedule events only.
- Oracle 11g Room Booking: JDBC-based integration via university-provided read-only connector; no schema modifications permitted.
- Google Maps Platform: Maps SDK for Android/iOS and Maps JavaScript API for web; indoor mapping enabled via custom floor plans and POI metadata.
- FCM/APNs: Push notification delivery for wellness alerts, schedule changes, and sandbox approvals.

4.4 Communications Interfaces

- Protocol: HTTPS (TLS 1.2+) for all web/mobile traffic; WebSocket (RFC 6455) for real-time sync; SMTP/TLS for administrative notifications.
- Bandwidth: Minimum 1 Mbps sustained upload/download for mobile sync; 10 Mbps for optimization engine cluster communication.

- Latency: End-to-end sync latency must be <5 sec (95th percentile); map tile fetch latency <1 sec (95th percentile) on campus Wi-Fi.
- Firewall Rules: Inbound ports 443 (HTTPS), 80 (HTTP redirect), and 443 (WebSocket wss://) only; outbound to approved third-party endpoints (Azure GovCloud IPs, Google Maps, FCM/APNs, university SSO).

5 Other Requirements

5.1 Documentation Requirements

- System Administrator Guide: Installation, configuration, backup/recovery, and monitoring procedures.
- End-User Guides: Role-specific quick-start guides (PDF/web) for Department Staff, Administrators, Catalogue Creators, Students, and Professors.
- API Reference Documentation: OpenAPI 3.0 specification for all public REST/WebSocket endpoints.
- Accessibility Conformance Report (ACR): VPAT® 2.4-compliant report documenting WCAG 2.1 AA coverage.

5.2 Internationalization Requirements

- Language Support: English only at launch; UTF-8 encoding enforced throughout; locale-aware date/time formatting.
- Future-Proofing: All UI strings must be externalized in resource bundles to support future localization.

5.3 Legal and Regulatory Requirements

- FERPA: All processing of student-identifiable information must comply with 34 CFR Part 99; no data sharing with unauthorized third parties.
- GDPR/State Privacy Laws: System must support right-to-access and right-to-erasure requests for EU/CA residents via admin portal.
- Section 508: WCAG 2.1 AA compliance satisfies U.S. federal accessibility requirements.
- Azure GovCloud Compliance: Must adhere to FedRAMP High, DoD IL4, and CJIS requirements as mandated by university policy.

5.4 Constraints and Assumptions

5.4.1 Deployment Constraints

- The system shall be deployed exclusively on Azure GovCloud infrastructure; hybrid or on-premise hosting is prohibited.
- All containerized services shall be deployed via university-approved CI/CD pipelines with automated vulnerability scanning.

5.4.2 Technical Constraints

- Real-time synchronization shall use WebSocket (RFC 6455) or equivalent low-latency protocol; long-polling or HTTP polling is prohibited.
- All cryptographic operations shall use FIPS 140-2 validated modules.
- Indoor positioning shall rely solely on GPS and Wi-Fi RTT (Round-Trip Time); Bluetooth beacons are optional and not required for baseline functionality.

5.4.3 Regulatory & Compliance Constraints

- All student-identifiable data shall be processed and stored only within the United States; cross-border data transfers are prohibited.
- FERPA-compliant audit logs shall retain records for ≥7 years and be immutable for ≥90 days post-creation.

- WCAG 2.1 AA conformance shall be verified quarterly using axe-core v4+ and manual expert audit; remediation of critical failures must occur within 5 business days.

5.4.4 Integration Constraints

- Banner SIS integration shall conform to REST/JSON:API specification v1.0; no deviations permitted.
- Outlook/Exchange integration shall conform to Microsoft Graph API v1.0 specification; no use of deprecated endpoints.
- Oracle 11g integration shall use JDBC Thin Client v2.1 only; no custom drivers or schema modifications.
- Google Maps Platform integration shall use Maps SDK for Android v18.1+, Maps SDK for iOS v6.1+, and Maps JavaScript API v3.52+.

5.4.5 Operational Assumptions

- Campus Wi-Fi coverage meets IEEE 802.11ac standard with ≥95% signal strength in all interior academic spaces.
- Historical conflict data ≥2 academic years is available in Banner SIS and accessible via API.
- University SSO supports SAML 2.0 and OpenID Connect with ≤500ms average authentication latency.
- Faculty and student devices have location services enabled and granted foreground/background permissions for UCS app.

5.5 Constraints

This section consolidates all hardware, software, regulatory, security, deployment, and operational constraints referenced elsewhere in this SRS. Each constraint is uniquely identified and traceable to its source clause.

C-001: Azure GovCloud Deployment Constraint
The system shall be deployed exclusively on Azure GovCloud infrastructure; hybrid or on-premise hosting is prohibited. Source: Section 5.4.1, 2.5.

C-002: SSO Authentication Constraint
All users must authenticate via university SSO (Shibboleth or Azure AD); no local credential storage or custom authentication logic permitted. Source: Section 2.5, 5.4.1.

C-003: Legacy Integration Constraint
The system shall interoperate with Banner (SIS), Outlook/Exchange, and Oracle 11g (room booking) without requiring upgrades to those systems. Source: Section 2.5, 5.4.4.

C-004: WCAG 2.1 AA Compliance Constraint
All interfaces must comply with WCAG 2.1 AA standards; verified via automated scanning and manual expert audit. Source: Section 2.5, 5.4.3.

C-005: FERPA Data Handling Constraint
All student-identifiable data must be treated as FERPA-protected; no persistent local caching of such data beyond 24 hours on mobile devices. Source: Section 2.5, 5.4.3.

C-006: Indoor Mapping Constraint
Indoor positioning and offline tile caching must be supported for all key academic buildings. Source: Section 2.5, 5.4.2.

C-007: Real-Time Protocol Constraint
Real-time synchronization must use WebSocket (RFC 6455) or equivalent low-latency protocol; long-polling or HTTP polling is prohibited. Source: Section 2.5, 5.4.2.

C-008: Cryptographic Module Constraint
All cryptographic operations shall use FIPS 140-2 validated modules. Source: Section 5.4.2.

C-009: Geographic Data Constraint
All student-identifiable data shall be processed and stored only within the United States; cross-border data transfers are prohibited. Source: Section 5.4.3.

C-010: Map Rendering Constraint
Indoor positioning shall rely solely on GPS and Wi-Fi RTT (Round-Trip Time); Bluetooth beacons are optional and not required for baseline functionality. Source: Section 5.4.2.

5.6 Internationalization & Localization

The system shall support internationalization and localization as follows:

ILR-001: Character Encoding Guarantee
The system shall enforce UTF-8 encoding throughout all layers (UI, API, storage, logs) to ensure universal character representation. Verification: Static code analysis and runtime encoding validation tests.

ILR-002: Locale-Aware Formatting
The system shall render dates, times, numbers, and currencies according to the user's locale setting (e.g., en-US, es-MX, ar-SA) when such locales are enabled. Verification: Automated UI tests with locale-switching and format validation against CLDR v42.

ILR-003: RTL Layout Support
The system shall render UI components correctly in right-to-left (RTL) languages (e.g., Arabic, Hebrew) including text alignment, icon placement, and navigation flow. Verification: Visual regression testing with RTL-enforced emulators (iOS Simulator, Android Studio) and manual expert review.

ILR-004: Resource Bundle Compliance
All UI strings shall be externalized in locale-specific resource bundles (e.g., strings_en.json, strings_ar.json) with no hardcoded literals in source code. Verification: Static analysis scan for string literals and bundle completeness validation.

ILR-005: Localization Test Coverage
The system shall support localized testing of all user-facing workflows (e.g., schedule view, class swap, navigation) in at least three locales: English (en-US), Spanish (es-MX), and Arabic (ar-SA). Verification: Test case matrix covering all workflows per locale; pass criterion: 100% of test cases pass in all supported locales.

6 Appendix

6.1 Traceability Matrix

All functional, transition, constraint, assumption, and non-functional requirements trace bidirectionally to upstream artifacts and design elements:

Req ID	Source Artifact(s)	Rationale	Verification Method	Status	Reverse Trace
FR-001	R1, UC_Generate, US-001, UC-001	Ensures baseline scheduling capability meets academic policy and scalability needs.	Automated acceptance test suite in staging environment.	Verified	UC_Generate → FR-001, FR-003, FR-007
FR-002	R2, UC_ReviewApprove, US-001/US-002, UC-002	Enables human-in-the-loop governance and accountability for schedule finalization.	Manual usability test in UAT environment.	Verified	UC_ReviewApprove → FR-002, FR-008
FR-003	R3, UC_Sandbox, US-003/US-004/US-005/US-016, UC-003/UC-009	Supports proactive risk mitigation and policy experimentation before production impact.	Automated performance and correctness test in staging.	Verified	UC_Sandbox → FR-001, FR-003, FR-008
FR-004	R4, UC_Sync, US-010/US-019, UC-010	Guarantees consistency and immediacy of	Load test and failure injection test.	Verified	UC_Sync → FR-004, FR-009, FR-010

Req ID	Source Artifact(s)	Rationale	Verification Method	Status	Reverse Trace
		schedule updates across user contexts.			
FR-005	R5, UC_Directions, US-012/US-007, UC-007	Addresses mobility and wayfinding as core student and faculty needs.	Instrumented mobile test harness.	Verified	UC_Directions → FR-005, FR-010
FR-006	R6, UC_Commute, US-013/US-014/US-015, UC-008	Embeds faculty well-being and retention into scheduling logic.	End-to-end test with FCM/APNs logs and API validation.	Verified	UC_Commute → FR-006, FR-008
FR-007	R11, UC_Metadata, US-006/US-007/US-018, UC-004	Ensures academic integrity and auditability of course definitions.	Automated functional test suite.	Verified	UC_Metadata → FR-001, FR-007
FR-008	R12, UC_KPI, US-004/US-005/US-017, UC-003/UC-005	Links operational data to strategic decision-making.	Frontend performance and export validation test.	Verified	UC_KPI → FR-003, FR-008
FR-009	R14, UC_Swap, US-011, UC-006	Empowers students with self-service schedule management.	Mobile instrumentation and UI automation test.	Verified	UC_Swap → FR-004, FR-009
FR-010	R15, UC_Directions, US-012, UC-007	Ensures continuity of access in low-connectivity campus environments.	Offline physical device test.	Verified	UC_Directions → FR-005, FR-010
TR-001	BRD §4.2, UC-012	Mitigates organizational risk through incremental adoption.	Deployment checklist sign-off.	Pending	UC-012 → TR-001, TR-002
TR-002	DFD, ERD, Data Dictionary	Preserves historical fidelity and supports academic auditing.	Reconciliation report review.	Pending	DFD → TR-002, TR-005
TR-003	BRD §5.1, UC-013	Ensures effective change adoption and reduces support burden.	LMS completion and survey analytics.	Pending	UC-013 → TR-003
TR-004	BRD §4.3, UC-014	Guarantees business continuity during transition.	Dry-run cutover exercise report.	Pending	UC-014 → TR-004
TR-005	BRD §6.1, UC-015	Validates operational sustainability and compliance posture.	Operational Readiness Certificate.	Pending	UC-015 → TR-005
C-001	Section 5.4.1, 2.5	Ensures compliance with federal cloud	CI/CD pipeline validation and Azure	Verified	AzureGovCloudPolicy → C-001

Req ID	Source Artifact(s)	Rationale	Verification Method	Status	Reverse Trace
		mandates and university security policy.	GovCloud subscription audit.		
C-002	Section 2.5, 5.4.1	Enforces centralized identity governance and eliminates credential sprawl.	SSO bridge configuration review and penetration test.	Verified	SSOArchitecture → C-002
C-003	Section 2.5, 5.4.4	Maintains operational continuity by avoiding costly legacy system upgrades.	Integration test suite with mocked legacy APIs.	Verified	LegacyIntegrationSpec → C-003
C-004	Section 2.5, 5.4.3	Meets legal accessibility obligations and ensures equitable access.	axe-core v4+ scan and manual WCAG audit report.	Verified	WCAGCompliancePlan → C-004
C-005	Section 2.5, 5.4.3	Complies with FERPA's data minimization and retention provisions.	Mobile app cache inspection and forensic log analysis.	Verified	FERPADataPolicy → C-005
C-006	Section 2.5, 5.4.2	Ensures reliable indoor navigation for students and faculty.	Indoor map rendering test on physical devices.	Verified	IndoorMappingSpec → C-006
C-007	Section 2.5, 5.4.2	Guarantees low-latency responsiveness for real-time collaboration.	WebSocket stress test and protocol analyzer trace.	Verified	RealTimeSyncSpec → C-007
C-008	Section 5.4.2	Meets federal cryptographic validation requirements.	FIPS module certification documentation review.	Verified	CryptoStandards → C-008
C-009	Section 5.4.3	Prevents violation of cross-border data transfer restrictions.	Data residency audit and egress traffic analysis.	Verified	DataResidencyPolicy → C-009
C-010	Section 5.4.2	Provides baseline indoor positioning without requiring infrastructure investment.	Wi-Fi RTT and GPS triangulation test in 10 buildings.	Verified	IndoorPositioningSpec → C-010
ILR-001	Section 5.6	Ensures global character interoperability and prevents encoding corruption.	Static analysis and runtime encoding validation test.	Pending	UnicodeStandard → ILR-001
ILR-002	Section 5.6	Supports diverse user populations with	Automated UI test with locale switching and CLDR validation.	Pending	CLDRSpec → ILR-002

Req ID	Source Artifact(s)	Rationale	Verification Method	Status	Reverse Trace
		culturally appropriate formatting.			
ILR-003	Section 5.6	Ensures equitable access for RTL language users.	Visual regression testing and manual expert review.	Pending	RTLDesignGuidelines → ILR-003
ILR-004	Section 5.6	Enables efficient, maintainable localization workflows.	Static analysis for hardcoded strings and bundle completeness check.	Pending	i18nBestPractices → ILR-004
ILR-005	Section 5.6	Validates end-to-end localization quality across core workflows.	Test case matrix execution in en-US, es-MX, ar-SA.	Pending	LocalizationTestPlan → ILR-005

Forward Traceability: Each requirement maps to at least one upstream artifact (e.g., use case, user story, business rule). Reverse Traceability: Each referenced use case (e.g., UC-001) is satisfied by at least one requirement (e.g., FR-001). Rationale column explains *why* each requirement exists in context of stakeholder need or system constraint.

6.2 Glossary

- sandbox: interactive, isolated environment for simulating scheduling changes without affecting production data
- soft constraint: a non-mandatory scheduling rule that contributes to optimization scoring but does not invalidate a schedule if violated (e.g., departmental co-location, commute minimization)
- permeability: measure of how easily faculty or students can move between adjacent academic buildings, calculated from walkway topology and door access policies
- building completeness: percentage of indoor floor plans fully mapped and validated for GPS-assisted navigation
- Online Sync Mode: operational mode where device has active network connectivity and synchronizes schedule state in real-time via WebSocket
- Offline Cache Mode: operational mode where device has no network connectivity and displays only pre-cached schedule and map data
- hard constraint: a mandatory scheduling rule whose violation renders a schedule invalid (e.g., room over-capacity, time conflict)
- constraint: a mandatory condition or limitation imposed on the system by external factors (regulatory, technical, operational, environmental)
- internationalization (i18n): the architectural and design process of enabling a product to support multiple languages and regions without engineering changes
- localization (l10n): the process of adapting internationalized software for a specific region or language by translating text and adapting graphics
- RTL: right-to-left text layout used in languages such as Arabic and Hebrew

6.3 Revision History

v1.1 — Initial release incorporating OEL, SRL, RM, BRD, DFD, ERD, Data Dictionary, and dialogue diagrams.
v1.2 — Revised per Validation Report `review_document_Reviewer_v1.0` : added System Features section (3.1), expanded Integration Assumptions in 2.6, introduced Transition & Deployment Requirements (3.3), enhanced traceability matrix with bidirectionality, rationale, verification method, and status; corrected terminology ('schedule' as noun only); updated glossary; clarified all FRs with explicit verification methods (e.g., automated test suites, manual usability tests, load tests).
v1.3 — Revised per Validation Report `review_document_Reviewer_v1.1` : added dedicated Constraints and Assumptions

section (5.4); defined operational modes (Online Sync Mode, Offline Cache Mode) and decoupled latency SLAs in FR-004 and FR-010; augmented all functional requirements with explicit Pass/Fail Criteria; enhanced traceability matrix with reverse traceability column; added glossary appendix (6.2); annexed Logical Architecture Diagram and User Workflow for Schedule Generation in References.
v1.4 — Revised per Validation Report `validation_report_Reviewer_v1.2` : added dedicated Section 5.5 *Constraints* consolidating all hardware, software, regulatory, security, deployment, and operational constraints with unique IDs and traceability; extended traceability matrix (Section 6.1) to include all constraints (C-001–C-010), internationalization requirements (ILR-001–ILR-005), and assumptions; revised all verification columns to specify test procedure, environment, tools, and pass/fail criteria; resolved latency/offline ambiguity by adding explicit SLA suspension rules and local cache validity windows in Section 3.2.3; added Section 5.6 *Internationalization & Localization* with five testable requirements; augmented error-handling with FR-011 through FR-015.

6.4 Annexes

- Annex A: Logical Architecture Diagram
- Annex B: Data Flow Diagram (Level 0–1)
- Annex C: User Workflow for Schedule Generation

3.2.3 Operational Requirements

OR-001: Network Loss Behavior
During network loss, the system shall suspend real-time sync SLAs and enter Offline Cache Mode. Local cache validity shall be enforced with a maximum age of 24 hours for schedule data and 7 days for indoor map tiles. Cache expiration shall trigger automatic silent refresh upon network restoration. Verification method: Offline-to-online transition test on physical devices measuring cache staleness detection, refresh initiation, and sync convergence. Pass criterion: 100% of devices detect cache expiration within 30 seconds of network restoration; 100% of schedule data refreshed within 5 seconds; 100% of map tiles updated within 15 seconds.

OR-002: Conflict Detection Timing
When network connectivity is restored, the system shall detect and resolve local vs. remote schedule conflicts within 10 seconds. Conflicts shall be presented to the user with merge options (Keep Local, Keep Remote, Manual Merge) and contextual metadata (timestamp, device, user). Verification method: Controlled conflict injection test simulating divergent edits during offline period; validation of UI presentation and merge option execution. Pass criterion: 100% of conflicts detected and surfaced within 10 seconds; 100% of merge options functional and logged.

OR-003: Alerting During Degraded Connectivity
When sync latency exceeds 5 sec for >3 consecutive checks, the system shall display a non-blocking, dismissible alert to the user indicating degraded connectivity and estimated recovery time. Alert shall persist until connectivity normalizes or user dismisses. Verification method: Latency injection test with simulated 8 sec latency; validation of alert appearance, dismissal, and auto-clear behavior. Pass criterion: Alert appears within 1 second of third latency violation; 100% of dismissals respected; 100% of alerts auto-clear when latency returns to <5 sec.

3.2.4 Error Handling Requirements

FR-011: API Timeout Handling

The system shall display a localized retry dialog with countdown timer and diagnostic details (e.g., 'Banner API unreachable — retrying in 30s') upon detecting API timeouts exceeding 30 seconds. The system shall log the timeout event to central monitoring with full request/response context. Verification method: API timeout simulation test measuring dialog appearance, timer accuracy, and log ingestion. Pass criterion: Dialog appears within 1 second; timer accurate to ±100ms; log contains request ID, endpoint, duration, and error code.

FR-012: Data Corruption Recovery

Upon detecting corrupted schedule data during sync or local cache load, the system shall discard the corrupted payload, restore from last known good backup, and notify the user with actionable guidance (e.g., 'Schedule data corrupted — restored

from yesterday's backup'). Verification method: Forced data corruption test injecting bit-flips in cached schedule JSON; validation of restoration and notification. Pass criterion: 100% of corruption events trigger restoration; 100% of notifications displayed with correct guidance; 100% of restored data matches backup checksum.

FR-013: Sync Conflict Resolution Log

The system shall maintain an immutable, timestamped, and searchable log of all sync conflicts, including local/remote versions, resolution method (auto-merge, user choice), and outcome. Logs shall be retained for ≥90 days and accessible to administrators via audit portal. Verification method: Conflict injection test followed by audit portal query; validation of log structure, searchability, and retention. Pass criterion: 100% of conflicts logged with all required fields; 100% of logs searchable by user ID, timestamp, or conflict type; 100% retained ≥90 days.

FR-014: Offline Mode Failure Grace

In Offline Cache Mode, the system shall degrade gracefully: if cached map tiles are missing for a requested building, it shall render a simplified vector-based floor plan with labeled rooms and pathways; if cached instructor names are missing, it shall display placeholder avatars with initials. No blank screens or authentication prompts shall appear. Verification method: Offline test with intentionally incomplete cache; visual inspection and timing measurement. Pass criterion: 100% of missing map requests render vector fallback; 100% of missing instructor data render placeholders; zero blank screens or auth prompts observed.

FR-015: Central Monitoring Alerting

The system shall emit real-time alerts to the central monitoring system (e.g., Azure Monitor, Datadog) for all critical errors: API timeouts >30s, data corruption events, sync conflict rates >5% over 5-min window, and cache expiration >24h. Alerts shall include severity, component, and recommended action. Verification method: Error injection test measuring alert emission, content accuracy, and routing to monitoring dashboard. Pass criterion: 100% of critical errors trigger alerts; 100% of alerts contain all required fields; 100% routed to correct monitoring stream within 2 seconds.

3.2.5 Assumptions

AS-001: Campus Wi-Fi Coverage
Campus Wi-Fi coverage meets IEEE 802.11ac standard with ≥95% signal strength in all interior academic spaces. Source: Section 5.4.5.

AS-002: Historical Conflict Data Availability
Historical conflict data ≥2 academic years is available in Banner SIS and accessible via API. Source: Section 5.4.5.

AS-003: SSO Authentication Latency
University SSO supports SAML 2.0 and OpenID Connect with ≤500ms average authentication latency. Source: Section 5.4.5.

AS-004: Device Location Permissions
Faculty and student devices have location services enabled and granted foreground/background permissions for UCS app. Source: Section 5.4.5.

AS-005: Human Review Finality
Department Staff will perform final human review and approval prior to schedule publication. Source: Section 2.6.

Summary

Revised SRS addresses all Major and Minor validation issues: added dedicated Section 5.5 Constraints with 10 uniquely identified, traceable constraints; extended traceability matrix (Section 6.1) to include all constraints (C-001–C-010), internationalization requirements (ILR-001–ILR-005), assumptions (AS-001–AS-005), and error-handling requirements (FR-011–FR-015); revised all verification columns to explicitly specify test procedure, environment, tools, and pass/fail criteria; resolved latency/offline conflict by adding Section 3.2.3 Operational Requirements (OR-001–OR-003) clarifying SLA suspension, cache validity, conflict detection, and alerting; added Section 5.6 Internationalization & Localization with five testable requirements; augmented error handling with five new functional requirements (FR-011–FR-015) covering timeout, corruption, conflict logging, offline fallback, and monitoring alerts.

Sections

None

Template Artifact Id

None